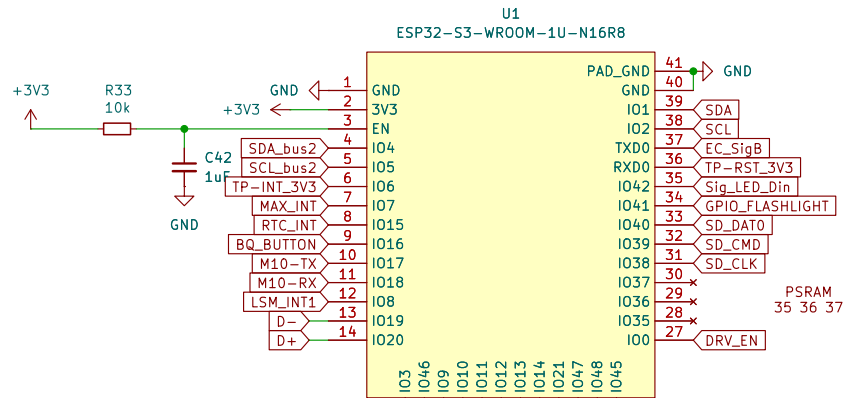
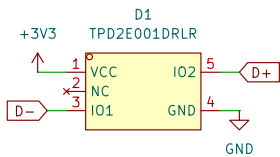


	1	2	3	4	5	6
A	1_ESP32-S3_Wroom File: 1_ESP32-S3_Wroom.kicad_sch	2_Power_Managment File: 2_Power_Managment.kicad_sch	3_Space_Time_Sensors File: 3_Space_Time_Sensors.kicad_sch			
B	4_Air_Sensors File: 4_Air_Sensors.kicad_sch	5_Lights File: 5_Lights.kicad_sch	6_PCB2_Skin_Sensors_LRA File: 6_PCB2_Skin_Sensors_LRA.kicad_sch			
C						
D						Ivan Porupski Aug2026(c) Sheet: / File: Kompic_Mk1b.kicad_sch Title: Kompic Mk1b Size: A4 Date: 2026-08-22 Rev: iv8.0 KiCad E.D.A. 10.0.5 Id: 1/7
	1	2	3	4	5	6

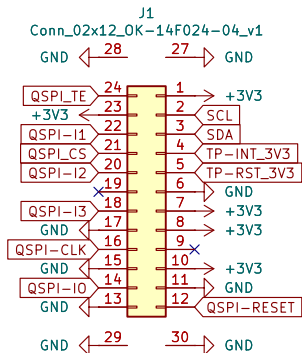
ESP32-S3 WROOM 1U N16R8



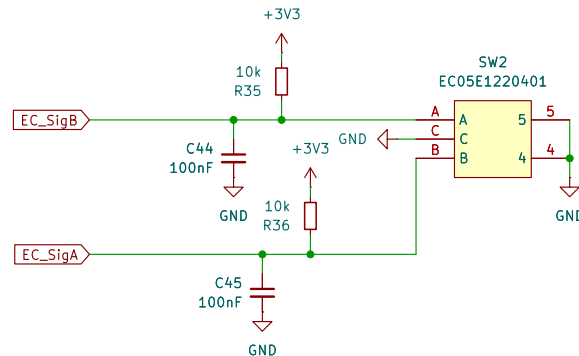
TVS Protection



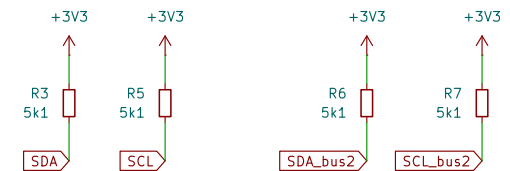
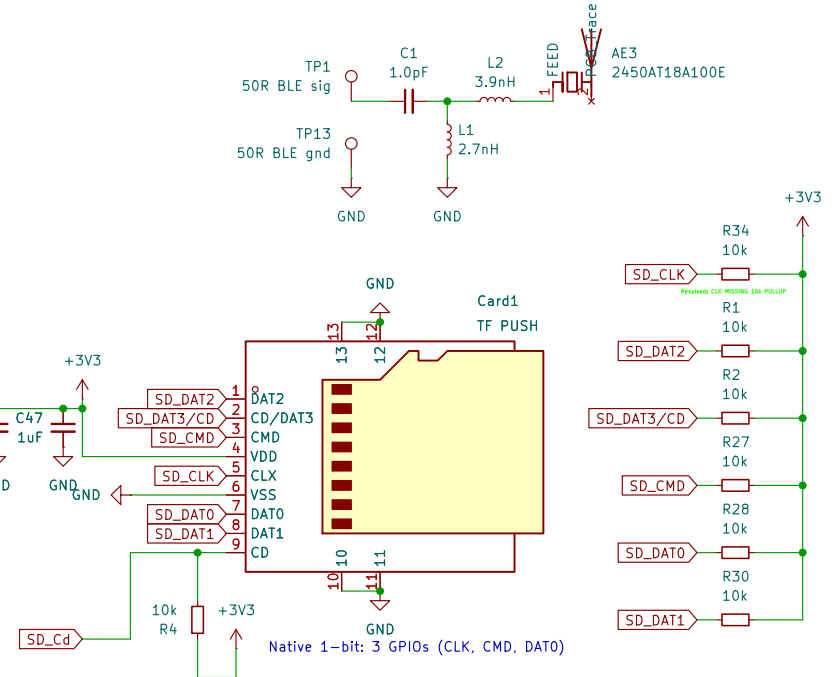
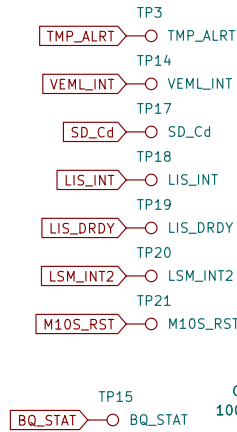
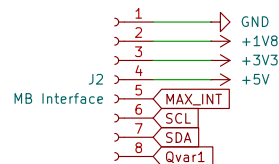
2.06-inch AMOLED (C05300AF-51+CST9217) OK-14F024-04 connector



Crown Encoder EC05 + Pushbutton + ECG



Daughterboard PCB MAX30101 + TMP117



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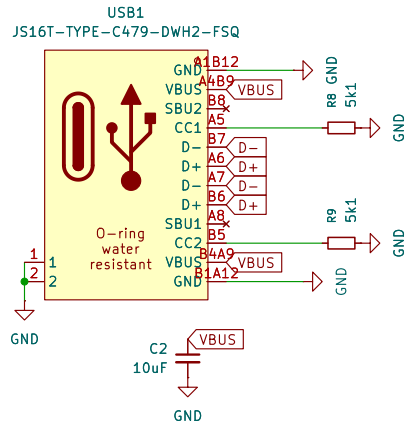
Sheet: /1_ESP32-S3_Wroom/
File: 1_ESP32-S3_Wroom.kicad_sch

Title:

Size: A4
Date: 2026-08-22
KiCad E.D.A. 10.0.5

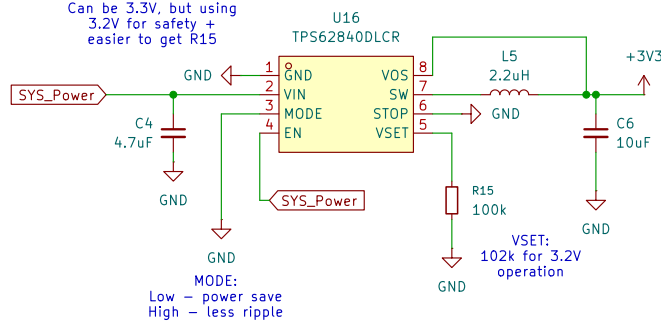
Rev: iv8.0
Id: 2/7

USB-C Connector Waterproof

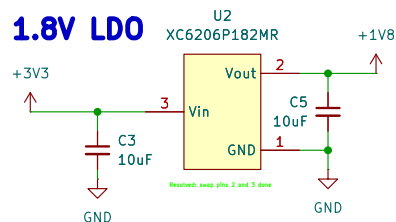


3.2V Buck

Can be 3.3V, but using 3.2V for safety + easier to get R15

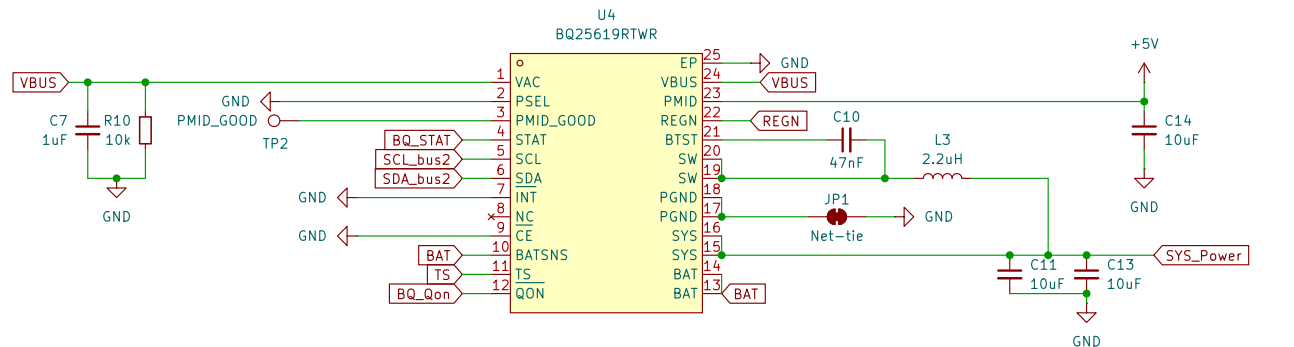


1.8V LDO

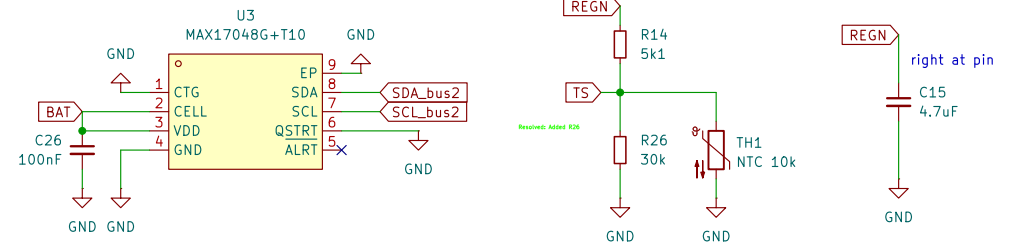
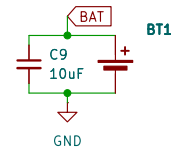


NOTE: library symbol NOT fixed!

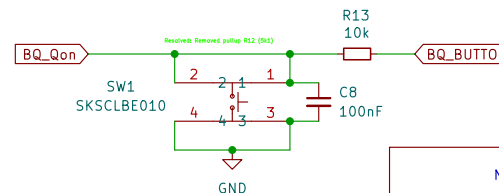
Power Managment (BQ25619)



LiPo 3.7V 380mAh



Tactile Switch GPIO+Wake



NTC 10k <https://jlcpcb.com/partdetail/MurataElectronics-NCP15XH103F03RC/C77131>

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Sheet: /2_Power_Managment/
File: 2_Power_Managment.kicad_sch

Title:

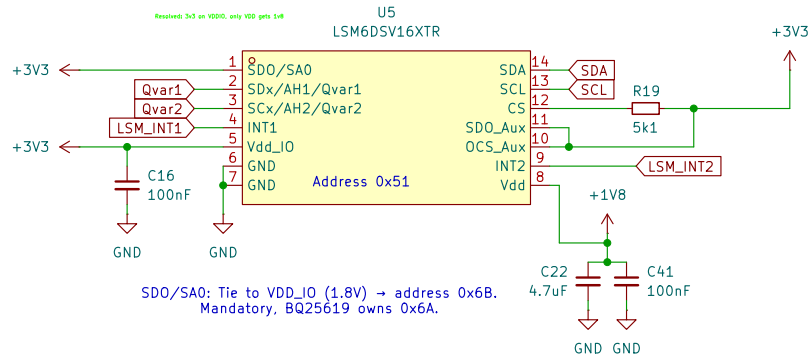
Size: A4 Date: 2026-08-22

KiCad E.D.A. 10.0.5

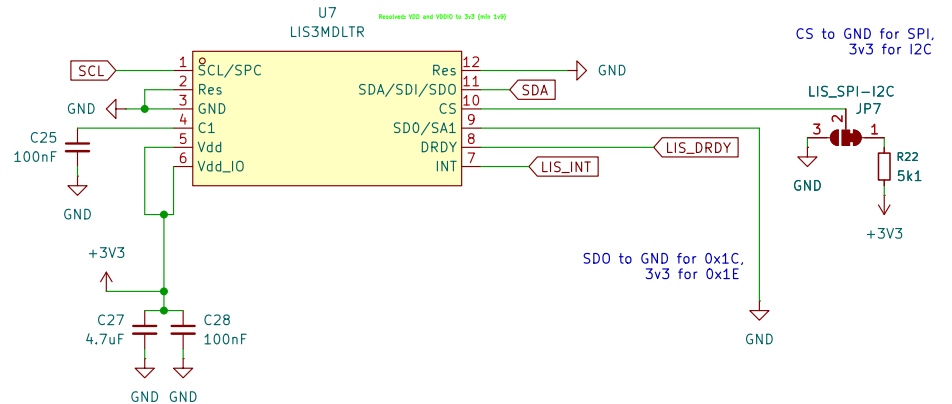
Rev: iv8.0

Id: 3/7

IMU (LSM6DSV)

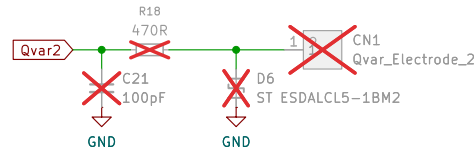


COMPASS (LIS3MDL)

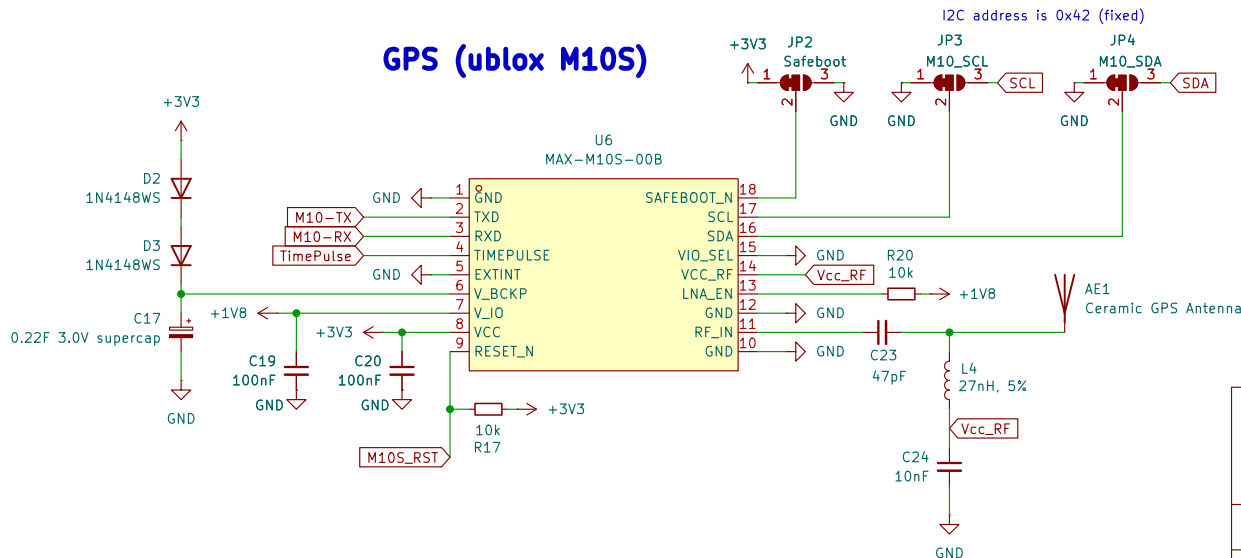


IMU (Qvar Touch)

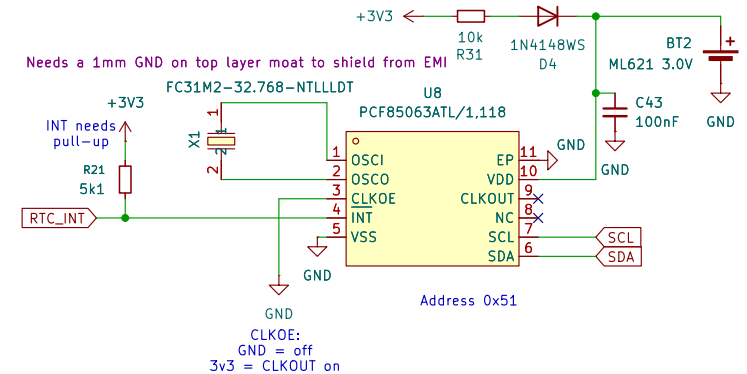
Electrode1 = Case Bump Electrode
Electrode2 = Crown Wheel Electrode
Keep TVS diodes close to pogo terminals.



GPS (ublox M10S)



RTC (PCF85063ATL/1,118)



Ivan Porupski Aug026(c)

Sheet: /3_Space_Time_Sensors/
File: 3_Space_Time_Sensors.kicad_sch

Title:

Size: A4
KiCad E.D.A. 10.0.5

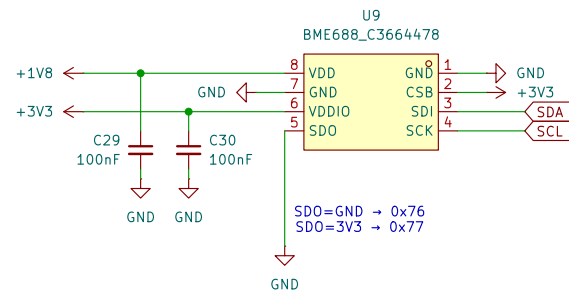
Date: 2026-08-22

Rev: iv8.0

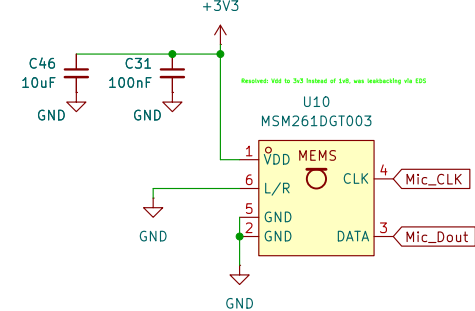
Id: 4/7

One hole, right side of case, ePTFE sticker over it,
mic and BME688 both routed to that cavity.

Air Quality (BME688)



MEMS Microphone



Ivan Porupski Aug2026(c)

Sheet: /4_Air_Sensors/
File: 4_Air_Sensors.kicad_sch

Title:

Size: A4 Date: 2026-08-22

KiCad E.D.A. 10.0.5

Rev: iv8.0

Id: 5/7

The circuit diagram shows the following connections:

- VDD (Pin 6):** Connected to +3V3.
- GND (Pin 1):** Connected to GND.
- SCL (Pin 5):** Connected to SCL.
- SDA (Pin 2):** Connected to SDA.
- INT (Pin 3):** Connected to VEML_INT.
- ADD (Pin 4):** Connected to GND.
- Capacitor C33:** A 100nF capacitor connected between VDD and GND.

Additional information provided in the diagram:

- U11 VEML6030
- ADD=GND → 0x10
- ADD=3V3 → 0x48

Resolved: Moved LED above Drain

GPIO_FLASHLIGHT

R32 3k

R24 10k

GND

+5V

R25 100R

LED2 TJ-S4008SW4TGLCCW-A5

U12 BSS138W_C890266

Id: 6/7

A



A



C

D

D



1



Id: 7/7